ARCHITECTURE REVIEW & STRATEGY

Kong Gateway LTS Upgrade

Migration Execution Blueprint: Transitioning seamlessly from 3.4 LTS → 3.10 LTS → 3.14 LTS

Upgrade Strategy at a Glance

0%

Downtime Target

Business Continuity & Reliability

By leveraging a dual-cluster Blue/Green Control Plane and Data Plane isolation deployment, API proxy traffic experiences zero disruption during schema and runtime transformations.

Automated decK CLI file conversions enable 100% entity schema validation across versions 3.4, 3.10, and 3.14 LTS releases prior to production cutover.

Architectural Phase Overview

Evaluating version milestones, security baseline changes, and operational prerequisites across the 3.x release series.

Major Technical & Breaking Changes

3.4 → 3.10 Migration Shift

OpenSSL 3.2 Upgrade: Enforces Security Level 2. RSA/DSA keys under 2048 bits and ECC keys under 224 bits are prohibited.

Deprecations: Cassandra DB support removed; Alpine and Ubuntu 18.04 builds discontinued.

Config Validation: Strict integer type checks applied to batch sizes and retry parameters.

3.10 → 3.14 Migration Shift

Default Protocols: Route default protocol is strictly HTTPS-only to safeguard endpoints by default.

OIDC & Auth Plugin Schemas: `consumer_claim` field formatted to array of arrays; headers claims unified.

Lua Security & Cryptography: Untrusted Lua sandbox defaults to strict; SHA1 deprecated in favor of SHA256.

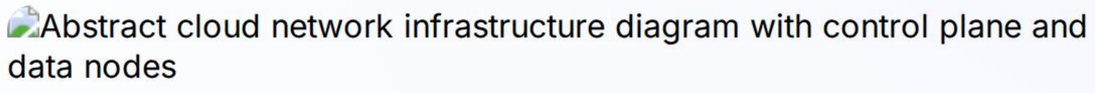
Feature & Breaking Changes Matrix

Architectural Vector	Kong 3.4 LTS	Kong 3.10 LTS	Kong 3.14 LTS (Target)
TLS / SSL Security Level	OpenSSL 1.1 / Level 1	OpenSSL 3.2 / Level 2	OpenSSL 3.2 / Level 2 + Strict TLS verify
Default Route Protocol	http, https	http, https	https (Secure by default)
OIDC Schema Format	Legacy Array of Strings	Legacy Array of Strings	Array of Arrays (`consumer_claims`)
Lua Execution Engine	Lax Sandbox	Lax Sandbox	Strict Sandbox (Network calls isolated)
Database Support	PostgreSQL / Cassandra	PostgreSQL only	PostgreSQL / DB-less / Konnect Hybrid

Dual-Cluster Architecture Blueprint

To mitigate runtime risk, we employ a **Dual-Cluster Blue/Green strategy** during Control Plane (CP) and Data Plane (DP) migration:

- **Control Plane Isolation:** Stand up a parallel 3.14 CP cluster backed by converted PostgreSQL schema.
- **Decoupled DP Migration:** Roll Data Plane worker nodes progressively without touching active 3.4 DP nodes.
- **DNS Traffic Shift:** Switch ingress load balancer traffic seamlessly once canary health checks pass.
- **Instant Fallback:** Revert DNS endpoints immediately back to Blue cluster in case of unforeseen edge anomalies.


Abstract cloud network infrastructure diagram with control plane and data nodes

Four-Phase Migration Timeline

Phase 1: Audit

Audit 3.4 plugin configs, custom Lua code, weak TLS certs, and decK schema rules.

Phase 2: decK Convert

Run decK 1.57+ convert (3.4 → 3.10 → 3.14). Validate YAML on staging environment.

Phase 3: Staging E2E

Deploy 3.14 Green CP & DP in Staging. Perform synthetic load & auth regression testing.

Phase 4: Prod Shift

Provision Prod Green cluster, execute canary traffic shifting, & decommission Blue nodes.

Infrastructure Prerequisites

Database & Storage

PostgreSQL 13+ required. Complete full database snapshots and verify keyring encryption keys prior to running decK migrations.

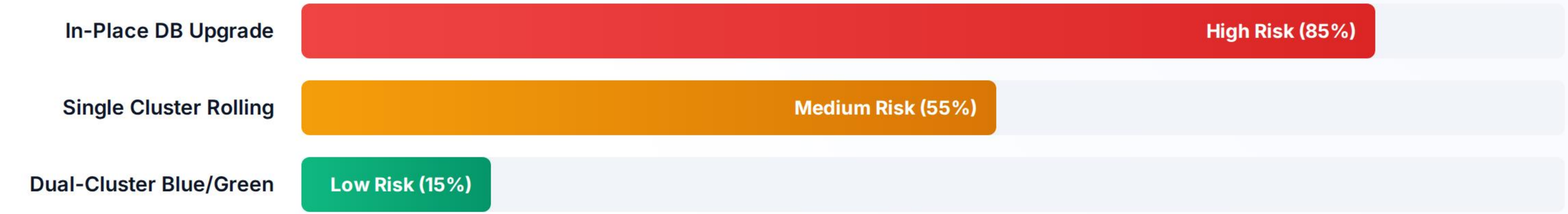
Compute Capacity Burst

Provision +100% temporary DP compute capacity to support dual-cluster coexistence during canary rollout window.

decK CLI Tooling

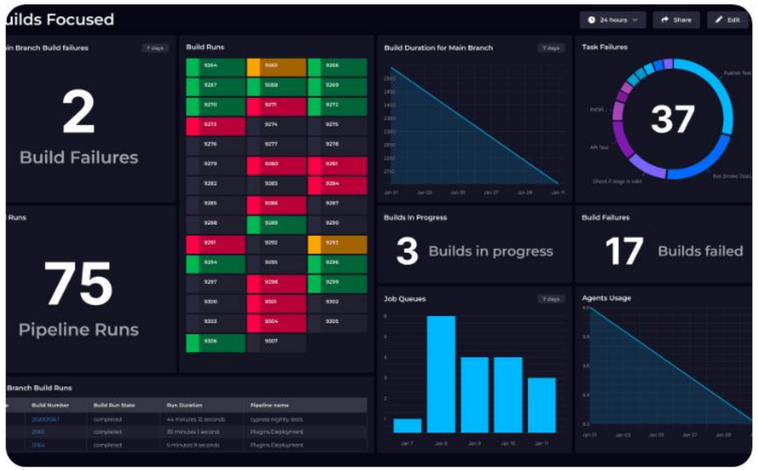
Upgrade decK CLI to v1.57.3 or higher across CI/CD runners to enable automated schema conversion flags (`--from 3.10 --to 3.14`).

Upgrade Strategy Risk Comparison



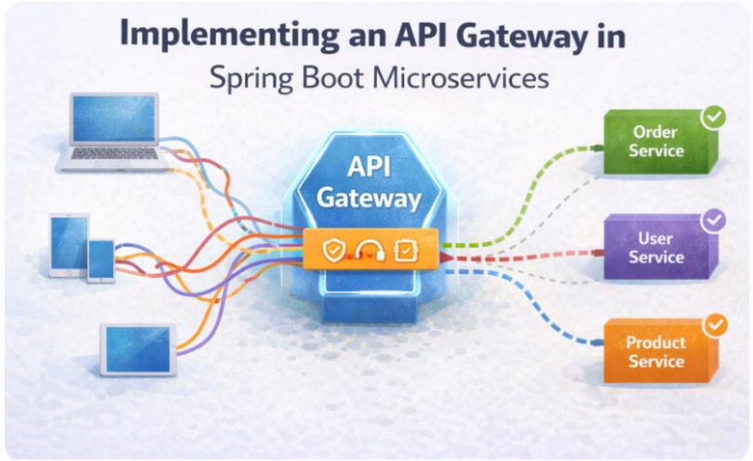
Dual-Cluster Blue/Green deployment minimizes operational downtime and risk by isolating state changes and providing instant rollback capabilities.

Core Team Responsibilities



DevOps & Infra Team

Infrastructure provisioning, Helm/Kubernetes manifests update, and Blue/Green ingress traffic routing control.



API Engineering

Custom Lua plugin refactoring, decK file conversion audits, and downstream API consumer testing.



Security & Compliance

Certificates audit for RSA 2048+ compliance, OIDC header claim validation, and keyring recovery testing.

Governance & Cutover Protocol

Strict Sign-off Gates

Every phase transition requires explicit engineering and business verification. Automated decK dry-runs must report zero schema warnings prior to cutover.

In the event of anomalous latency or error rate spikes exceeding 0.01%, traffic automatically reverts to the 3.4 Blue cluster within 30 seconds.



Executive Q&A & Next Steps

Thank you. We welcome questions and discussion regarding the architectural roadmap.

Infrastructure Architecture Team | Enterprise Gateway Taskforce

Image Sources



https://kubernetes.io/images/docs/kubernetes-cluster-architecture.svg

Source: kubernetes.io



https://squaredup.com/_next/image/?url=https%3A%2F%2Fcdn.sanity.io%2Fimages%2Fz7wg6mcy%2Fproduction-2025%2Fc55d5bc1a28449412300df2c17a2f7e4218623d2-1920×1080.png%3Fq%3D100%26fit%3Dmax%26auto%3Dformat&w=3840&q=75

Source: squaredup.com



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Source: medium.com



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